

Low-Level Design (LLD)

Designer Catalogue & Inspiration Platform

1. Overview

This Low-Level Design (LLD) details the **internal system design** for a **fullstack Django (server-rendered)** application, including: - Django apps & models (ORM) - Views (function/class-based) - URLs & templates - Forms, permissions & validations

Tech Stack Assumption: - Django (MTV architecture) - Django Templates - PostgreSQL - Session-based Authentication

2. Authentication & Authorization

2.1 Auth Flow

- Django session-based authentication
- Login / Logout / Register views

2.2 User Roles

- USER (default)
- DESIGNER
- ADMIN (staff/superuser)

Authorization handled using: - @login_required - User flags (is_designer, is_staff) - Custom decorators / mixins

3. Core Database Models

3.1 User (Django Default)

```
User
- id (UUID)
- username
- email
- password
- is_designer (bool)
```

- is_active
- created_at

3.2 DesignerProfile

```
DesignerProfile
- id (UUID)
- user (OneToOne → User)
- brand_name
- bio
- profile_image
- location
- verified (bool)
- created_at
```

3.3 Catalogue

```
Catalogue
- id (UUID)
- designer (FK → DesignerProfile)
- title
- description
- cover_image
- is_published
- created_at
- updated_at
```

3.4 Design

```
Design
- id (UUID)
- catalogue (FK → Catalogue)
- title
- description
- media (image/video)
- price
- contact_for_price (bool)
- tags (ManyToMany → Tag)
- created_at
```

3.5 Tag

```
Tag
- id (UUID)
- name
- slug
```

3.6 Board

```
Board
- id (UUID)
- user (FK  User)
- title
- is_private
- created_at
```

3.7 SavedItem

```
SavedItem
- id (UUID)
- user (FK  User)
- design (FK  Design, null=True)
- catalogue (FK  Catalogue, null=True)
- board (FK  Board)
- created_at
```

3.8 Message

```
Message
- id (UUID)
- sender (FK  User)
- receiver (FK  User)
- content
- is_read
- created_at
```

3.9 Subscription

```
Subscription
- id (UUID)
- designer (FK  DesignerProfile)
- plan (FREE / PRO / PREMIUM)
- start_date
- end_date
- is_active
```

4. URL Routes & Views (No APIs)

4.1 Authentication Routes

URL	View	Description
/register/	register_view	User registration
/login/	login_view	Login
/logout/	logout_view	Logout

4.2 Designer Routes

URL	View	Description
/designer/apply/	apply_designer	Become designer
/designer/profile/	designer_profile	View own profile
/designer/profile/edit/	edit_designer_profile	Edit profile

4.3 Catalogue Routes

URL	View	Description
/catalogues/	catalogue_list	List catalogues
/catalogue/create/	catalogue_create	Create catalogue
/catalogue/<id>/	catalogue_detail	Catalogue detail
/catalogue/<id>/edit/	catalogue_update	Update catalogue

URL	View	Description
/catalogue/<id>/delete/	catalogue_delete	Delete catalogue

4.4 Design Routes

URL	View	Description
/design/create/<catalogue_id>/	design_create	Upload design
/design/<id>/	design_detail	View design
/design/<id>/edit/	design_update	Update design
/design/<id>/delete/	design_delete	Delete design

4.5 Feed & Discovery

URL	View	Description
/	home_feed	Main inspiration feed
/trending/	trending_feed	Trending designs

4.6 Boards & Save

URL	View	Description
/boards/	board_list	User boards
/board/create/	board_create	Create board
/save/design/<id>/	save_design	Save design
/save/catalogue/<id>/	save_catalogue	Save catalogue

4.7 Messaging

URL	View	Description
/messages/	inbox	Inbox
/messages/<user_id>/	chat_view	Chat with designer

4.8 Search

URL	View	Description
/search/	search_view	Search designs/catalogues

5. Validation & Business Rules

& Business Rules

- Only designers can create catalogues/designs
- Catalogue must exist before design upload
- Free plan limits number of catalogues
- Admin approval for featured catalogues

6. Error Handling

- Standard HTTP status codes
- Consistent error response format

```
{  
  "error": "Permission denied"  
}
```

7. Performance Considerations

- Pagination on all list APIs
- Redis caching for feed
- Async media processing

8. Security Considerations

- JWT expiration & refresh
- Input validation
- File upload limits
- Rate limiting on auth & messaging

9. MVP Boundary

Included: - Core CRUD APIs - Feed - Boards - Messaging

Excluded (Phase 2): - Payments - AI recommendations - Push notifications

End of LLD Document